

Supplementary Material

Do AI chatbots find what experts would? Effects of model, user role, and sample size on study retrieval for medical questions

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Repository: <https://github.com/QingfangLiu/llm-evidence-retrieval-bias>

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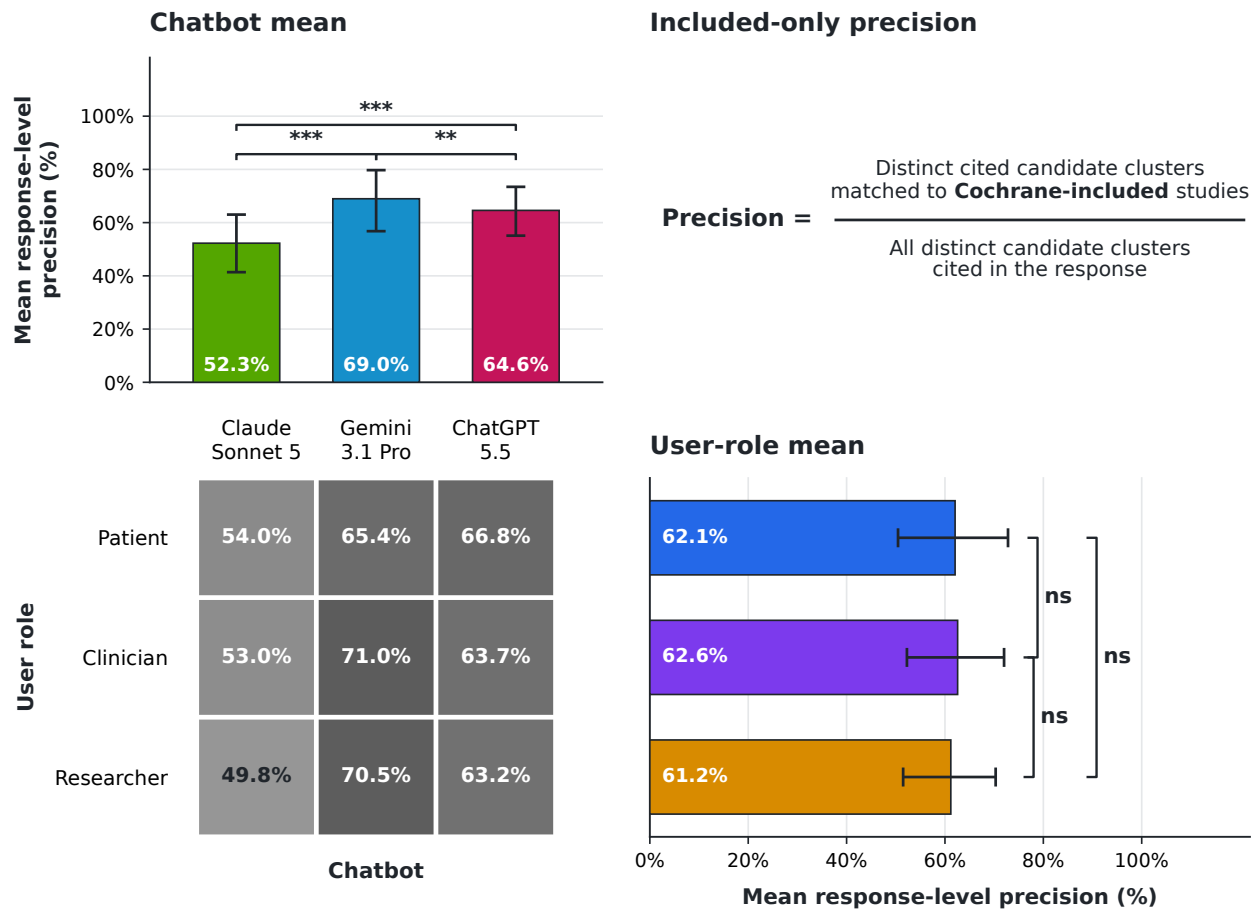


Figure S1: **Included-study precision by chatbot and user role.** For each response, precision is the number of distinct cited candidate studies matched to Cochrane-included studies divided by all distinct candidate studies cited. The heatmap shows means across 80 responses per chatbot–role combination; the bars show means across 240 responses per chatbot or role. Each response is weighted equally. Error bars are 95% review-clustered bootstrap confidence intervals. Brackets show Holm-adjusted, two-sided blocked permutation tests (50,000 permutations per pair); ** $p < 0.01$, *** $p < 0.001$, and ns $p \geq 0.05$.

Supplementary Tables

Table S1: **Examples of coded Cochrane-review mentions in chatbot responses.** Examples illustrate the four observed combinations of review identity and review role. Quoted text is reproduced verbatim from the saved chatbot responses.

Review identity	Review role	Example excerpt	Source
Matching 2026 review	Used to identify or verify studies	“Several sources I found were reviews (e.g., the Fransen 2010 hip OA meta-analysis, the 2026 Cochrane update, and various systematic reviews) — these were excluded per your criteria but were useful for identifying the primary trials embedded within them, which are listed below.”	Claude, CD007912, researcher, repetition 2
Matching 2026 review	Mentioned without a stated study-identification role	“For reference, a 2026 Cochrane systematic review (Larvie DY et al., <i>Vitamin B12 supplementation for growth, development, and cognition in children</i>) exists covering this exact question — I excluded it per your request, but its reference list is a useful cross-check for completeness if you want me to mine it further for additional primary trials.”	Claude, CD015264, clinician, repetition 1
Earlier or edition-unspecified review	Used to identify or verify studies	“Note: a few of the earlier trials (Walti, Bevilacqua, Egberts, Iarukova) I sourced from citation lists in the Cochrane review rather than the primary articles directly”	Claude, CD000510, clinician, repetition 3
Earlier or edition-unspecified review	Mentioned without a stated study-identification role	“Here’s what the primary-study evidence shows on prophylactic vs. selective surfactant, based only on individual trials (not the Cochrane reviews, guidelines, or narrative articles that came up in search).”	Claude, CD000510, patient, repetition 1